

Sarvesh Prasanth Venkatesan

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EDUCATION

University of Southern California <i>M.S. in Electrical and Computer Engineering - Machine Learning</i>	Los Angeles, CA 2025-2027
Vellore Institute of Technology <i>B.Tech. in Electronics and Communication Engineering GPA: 3.7/4.0</i>	Vellore, India 2019 - 2023

EXPERIENCE

Stochastic Robotics Lab Indian Institute of Science <i>Project Associate Advisor: Prof. Shishir Kolathaya</i>	Jul 2024 – Aug 2025 Bengaluru, India
- Designed a recurrent-transformer-based generalist quadruped policy leveraging offline imitation learning to fuse diverse locomotion skills, achieving zero-shot skill transfer across 5+ unseen quadruped morphologies. [Webpage] - Reduced sim-to-real gap on 70 kg Stoch5 quadruped by implementing novel <i>Ground-Reaction Force(GRF)</i> and <i>adaptive Raibert-inspired rewards</i>, improving push recovery by 75–98% on 30° low-traction inclines. [Videos] - Explored visual locomotion policies using deep reinforcement learning to enable locomotion on challenging terrains.	
Mobile Robotics Lab Indian Institute of Science <i>Project Associate Advisors: Prof. Debashish Ghose</i>	Sep 2023 – Jun 2024 Bengaluru, India
- Developed a heterogeneous multi-agent system for large-scale agricultural monitoring. [Video] - Collaborated on LocalizeSORT, a multi-object tracking framework with world-coordinate association, achieving +15% HOTA and +30% prediction accuracy. [Paper] - Built a vision-based autonomy stack for aerial and quadruped robot using cuVSLAM.	
Centre for Non-Destructive Evaluation Indian Institute of Technology Madras <i>Intern Advisors: Prof. Prabhu Rajagopal</i>	Sep 2022 – Jul 2023 Chennai, India
- Designed and developed software for a Shopping Assistance Robot. [Video] - Implemented RRT-based motion planning with vision for manipulation and navigation for automated grocery pickup tasks.	

PUBLICATIONS

- Srihari Vemuru, Kumar Ankit, **Sarvesh Prasanth Venkatesan**, Debashish Ghose, Shishir Kolathaya. [Yield Prediction and Counting Using World Coordinates](#). Under Review at *Computers and Electronics in Agriculture*.
- Narayanan PP, **Sarvesh Prasanth Venkatesan**, Srinivas Kantha Reddy, Shishir Kolathaya. [GRoQ-LoCO: Generalist and Robot-agnostic Quadruped Locomotion Control using Offline Datasets](#). *arXiv*.

PROJECTS

Vaathi - AI Tutor (Github) <i>Good Vibes Only Hackathon</i>
- Built an AI Tutor with multimodal pipelines for text, audio, and document understanding using OCR, semantic search, and LLM reasoning. - Developed multi-backend LLM framework (OpenAI GPT-4o-mini with Ollama fallback) featuring prompt chaining, context window optimization, in-memory caching, and exponential-backoff retries. - Implemented real-time voice interaction loop using Whisper ASR and FastTTS v2 with low-latency audio orchestration, enabling continuous, context-aware dialogue generation.
Humanoid Control (Github)
- Implemented a visual humanoid locomotion policy based on this paper. [Video] - Training Nvidia GR00T N1.5 for Unitree G1 on language-guided pick-and-place in simulation.

SKILLS

Programming:	Python, C++, C, R, Verilog.
ML/Robotics:	ROS, MoveIt, Nvidia Isaac, MuJoCo, PX4, ArduPilot, PyTorch, Tensorflow, JAX, OpenCV.
Software:	Docker, Git, AWS, Fusion360, MATLAB, LaTeX, Ubuntu Linux.
Hardware:	Unitree Go1, UR5, Husky, PiPER, Nvidia Jetson, Intel RealSense, Moteus.